

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.11: Internet Hardware (Modems, PSTN)

Total Questions: 3

Mark Schemes begin on Page 4

Question 1 | Source: 9618_s25_qp_11 (Q2.c)

- 2 Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

(c) The video conference is accessed over the internet.

Complete the table by stating how modems **and** dedicated lines are used when data is transmitted over the internet.

Hardware	Use
modems	
dedicated lines	

[2]

Question 2 | Source: 9618_s25_qp_12 (Q6.b)

- 6 A company has multiple sites in different cities.
The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(b) The smartphones use the cell phone network to connect to the WAN.

Explain how data is transmitted using the cell phone network.

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[4]

Question 3 | Source: 9618_s24_qp_12 (Q3.c.ii)

3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(c) Data transmitted on the internet passes through multiple different systems.

(ii) Describe the role of the PSTN (Public Switched Telephone Network) in the transmission of data through the internet.

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_11 (Q2.c)

2(c)	1 mark each modems: - Converts digital data into analogue for transmission down phone lines // Converts analogue data into digital after transmission down phone lines dedicated lines: - Used to provide a direct / private connection which therefore provides faster transmission	2
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Mark Scheme for Question 2 | Source: 9618_s25_ms_12 (Q6.b)

6(b)	1 mark each to max 4 e.g. - Land is split into cells designed for maximum line of sight - Each cell has a tower with an antenna which receives and transmits data - Data is transmitted between the tower and the phone - Data transmission is wireless using low power radio signals/frequencies - Multiple devices can communicate simultaneously with the same tower	4
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Mark Scheme for Question 3 | Source: 9618_s24_ms_12 (Q3.c.ii)

3(c)(ii)	1 mark each to max 2: - The PSTN consists of many different types of communication lines ... therefore the digital data may need to be converted into a different form/analogue signal - Data is transmitted in both directions at the same time // duplex data transmission - Using a PSTN the communication passes through different switching centres/ISPs	2
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